Jeronimo Ruiz Fernandez

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EDUCATION

Worcester Polytechnic Institute Graduation: May 2027

B.S. in Robotics Engineering and B.S. in Computer Science GPA: 3.6/4.0, Dean's List

Coursework: Unified Robotics I–IV; Machine Learning; Deep Learning; AI for Robotics; Software Engineering

WORK EXPERIENCE

Argano, Netlogistik SaaS team MCP, SQL, Snowflake, Node.js May 2026 - August 2026

Software Engineering Intern Remote

- **Strengthened the SaaS offering for five enterprise clients (~$1.2M ACV) by migrating** a single-shot JSON-parsing chatbot **to an agentic multi-step tool-use framework** and **building an LLM-as-a-Judge evaluation pipeline** driven by client feedback signals.
- **Built a RAG pipeline on Azure AI Search**, embedding a ~547-chunk logistics corpus and integrating retrieval as a tool in the service's function-calling agent loop.

Efficient Learning and Planning for Intelligent Systems Lab ROS2, Python, RealSense, HDBSCAN, UR10 September 2024 - Present

Research Assistant WPI

- **Built an end-to-end placement selection pipeline for a UR10 robotic arm** and Intel RealSense setup **for the ManipulationNet 'Text to Placement Task' competition**. It consumes an instance segmentation mask and an aligned depth frame and returns a collision-free (x, y, z) placement via a ROS2 service interface.
- **Enabled arm functionality in complex object-scattered environments** by performing scene understanding, depth normalization, HDBSCAN clustering in (x, y, z), and kernel convolution of the obstacle mask.

Horizon Surgical Systems October 2025

Apprenticeship Santa Monica, CA

- **Enhanced the Polaris™ robotic system reliability** by investigating & reproducing flaws found in cataract extraction trials and contributing to the hardware fix proposal.
- **Improved the robotic platform's surgical precision** with unimpeded trajectory following and generation by helping with debugging and testing trajectory regressions in multiple automated surgical steps.

PROJECTS

Autonomous Frontier Exploration on TurtleBot3 (ROS 2 Jazzy, Python) March 2026 - May 2026

- **Built a ROS 2 autonomy stack enabling** a physical TurtleBot3 **to explore and map unknown environments**, implementing vectorized frontier detection, A* planning, and pure pursuit control.

Automated Coating System Sponsor: Saint-Gobain Ongoing

- **Developing a UR10 system to automate Mayer-rod coating** with cleaning verification and tool management; **built a MuJoCo digital twin** for IK-based stroke trajectory execution.

SKILLS

Programming: Python, C++, C, MATLAB, Java, TypeScript | **Robotics:** ROS2, RViz, MoveIt, MuJoCo, Linux, PyTorch, OpenCV | **Perception:** Depth Sensing | **Embedded Systems:** Microcontrollers | **CAD:** SolidWorks